Zero-Prior Diagnostic Agent for Neutron Measurement Data with Manual-Grounded RAG: System Design and Retrieval Robustness Evaluation

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Abstract

Nuclear power plant neutron instrumentation channels produce large volumes of sparsely labeled time-series data, while operational diagnosis still relies on technical manuals such as IAEA guidance. Pure rule-based pipelines struggle to generalize, whereas large language models alone risk ungrounded recommendations. This paper presents an end-to-end framework that (i) uses a zero-prior tool-using agent to profile neutron-current CSV records and generate condition reports, (ii) compresses reports into structured JSON to drive multi-stage retrieval-augmented generation (RAG) over nuclear I&C manuals with citable evidence, and (iii) systematically evaluates dense retrieval under controlled expansion of a same-domain non-core supplement around a fixed IAEA core library, together with query language, chunking, and relevance criteria. Experiments show that cosine scores can rise while core-document purity collapses as the core–non-core ratio changes, that Chinese short queries fail much earlier than English ones, and that chunk-size/overlap effects are secondary to language and query formulation. The results motivate report-driven, terminology-rich query construction and careful governance of core versus supplementary manuals for knowledge-grounded nuclear diagnostics.

Keywords: neutron instrumentation, online monitoring, large language model agent, retrieval-augmented generation, nuclear I&C manuals, retrieval robustness

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1. Introduction

Nuclear power plants (NPPs) continuously acquire neutron and instrument-channel measurements for surveillance and condition monitoring. Ex-core and in-core detectors yield high-volume time series that support power surveillance, sensor health assessment, and early anomaly detection, yet the resulting records are typically sparsely labeled. In practice, day-to-day diagnosis still depends on authoritative technical manuals, including IAEA guidance on online monitoring (OLM) for instrument channels and for process and component condition monitoring (International Atomic Energy Agency, 2008a,b). Hashemian (Hashemian, 2011) reviewed OLM technologies that extend calibration intervals and strengthen channel performance assessment. Beyond calibration surveillance, reactor-physics studies treat neutron-noise analysis as a non-intrusive probe of vibrations, flow anomalies, and core dynamics (Pázsit and Demazière, 2010; Demazière et al., 2022), while data-driven methods have automated validation of neutron-detector operation and noise-signal integrity under constrained detector sets (Tambouratzis et al., 2020; Tambouratzis, 2020). Deep-learning fault diagnosis for NPPs continues to improve accuracy on labeled accident and fault scenarios (Elbordany et al., 2024); however, such pipelines usually assume fixed feature schemas or predefined fault libraries. Exploratory diagnosis of neutron-current CSV streams often lacks gold-standard fault tags, so operators must still map observed spikes, dropouts, and collinear channel groups to clauses in I&C manuals. Rule-only workflows are brittle across diverse plant conditions, whereas large language models (LLMs) used alone may produce fluent but ungrounded recommendations. These observations motivate language-mediated assistants that observe measurement data with weak priors and then ground interpretations in citable manuals.

Retrieval-augmented generation (RAG) (Lewis et al., 2020) addresses part of this grounding problem by conditioning generation on passages retrieved from an external corpus. In nuclear applications, the corpus is typically regulatory text, plant technical specifications, work orders, or design and safety manuals rather than open-domain web content. Idaho National Laboratory studies have shown that RAG can improve diagnostic question answering and predictive-maintenance explanations for NPP systems, while emphasizing factuality evaluation because nuclear coverage in pretraining remains sparse (Reeves et al., 2024; Lin et al., 2025). Work on power-uprate document interpretation further illustrates both the utility of RAG over thousands of technical pages and recurring failure modes, including incorrect retrieval, misinterpretation of retrieved text, reliance on parametric knowledge outside the evidence, hallucination, and refusal (Mapes et al., 2025). Agentic, citation-enforcing frameworks such as RADIANT-LLM target local, provenance-tracked decision support for safety-critical nuclear workflows and report marked reductions in hallucination relative to ungrounded commercial chatbots (Ndum et al., 2026). Taken together, these studies indicate that manual-grounded RAG is a practical route to more trustworthy nuclear language assistance. Nevertheless, most nuclear RAG demonstrations begin from a user question about documents (procedures, technical specifications, or maintenance records). Far fewer systems couple retrieval to an upstream analysis of neutron measurement CSV data, where the query itself must be constructed from observed channel phenomena rather than from free-text operator prompts.

Tool-using agents provide a complementary mechanism for the measurement side of the pipeline. ReAct-style agents (Yao et al., 2023) interleave reasoning with tool calls, enabling iterative observation–action loops over local analyzers and external interfaces. For neutron-current diagnosis, this paradigm suggests a persistent workspace blackboard together with specialized probes—profiling, zero-prior statistical discovery, anomaly and regime segmentation, lag checks, and spatial snapshots—before any manual retrieval is invoked. On the knowledge side, practical RAG stacks (Liu, 2022) rely on chunking, dense embedding, and top-k retrieval, while evaluation frameworks such as RAGAs (Es et al., 2024) emphasize faithfulness and context relevance of generated answers. Nuclear RAG studies already caution that retrieval errors can propagate into unsafe advice (Mapes et al., 2025), and corpus-scaling experiments show that knowledge-base growth alters context precision (Ndum et al., 2026). What remains under-characterized for neutron-manual corpora is the relationship between a compact authoritative *core* library and a larger same-domain *non-core* supplement: as the core–non-core ratio changes, cosine similarity scores may rise while core-document purity collapses, especially under short Chinese versus English queries and under strict chunk identity versus document-level relevance. General RAG benchmarks rarely stress bilingual nuclear terminology or core-versus-supplement purity under controlled library expansion.

Against this background, the present work addresses three coupled challenges: (1) how to extract spikes, dropouts, collinearity or redundancy, and regime changes from neutron-current CSV data with minimal prior assump-

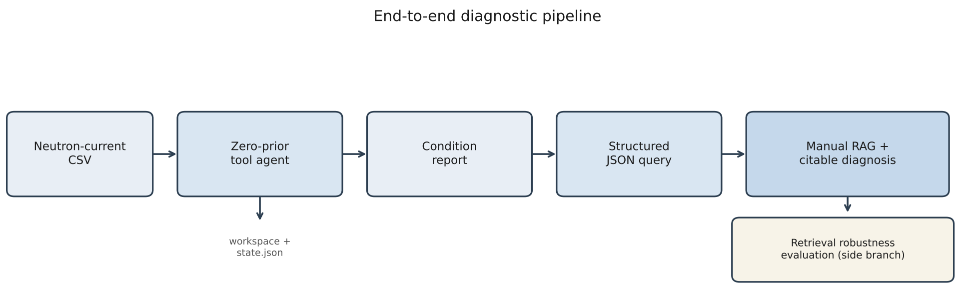

Figure 1: End-to-end pipeline: neutron-current CSV is analyzed by a zero-prior tool-using agent that maintains a workspace blackboard and produces a condition report; the report is compressed into structured JSON to drive manual-grounded RAG with citable diagnoses. A side branch evaluates retrieval robustness under same-domain corpus expansion.

tions; (2) how to align observed phenomena with manual clauses while retaining citable provenance; and (3) whether dense retrieval over same-domain technical corpora remains reliable after corpus expansion, particularly under Chinese versus English query formulations. The main contributions are as follows. First, we develop a zero-prior, tool-using diagnostic agent with a workspace blackboard for neutron-current analysis. Second, we introduce a report-driven multi-stage RAG protocol that converts condition reports into structured JSON queries, retrieves passages from IAEA-centered nuclear I&C manuals, and produces expert-style diagnoses with explicit evidence identifiers. Third, we conduct a same-domain retrieval-robustness study that quantifies corpus dilution, language effects, chunking choices, and the gap between strict chunk matching and document-level relevance. Figure 1 summarizes the end-to-end pipeline.

The remainder of this paper is organized as follows. Section 2 presents the data resources and system design, including the diagnostic agent, manual corpus indexing, and report-driven RAG diagnosis. Section 3 describes the evaluation protocol for retrieval robustness. Section 4 reports experimental results and discussion. Section 5 concludes the paper and outlines future work.

2. System design

2.1. Data description

Two complementary resources are used throughout this study: plant neutron-current measurements for agent-side diagnosis, and a curated nuclear

I&C manual corpus for manual-grounded RAG and retrieval evaluation.

Neutron-current measurement archive.. The underlying measurement archive is organized into four groups of multi-channel neutron-current CSV records (hereafter Groups A1, A2, B1, and B2), corresponding to different instrumentation banks in the same plant campaign. In total, the archive contains 42 series files, as summarized in table 1.

Table 1: Composition of the neutron-current measurement archive.

Group	Number of series	Channels per series	Columns per series
A1	11	7	8
A2	11	7	8
B1	10	7	8
B2	10	7	8
Total	42	—	—

Each series stores a time column together with seven neutron-current channels (eight columns overall); channel identifiers follow plant tagging conventions (e.g., `RIC...MA_VER`). Within each group, the seven detectors are arranged in a circular (circumferential) layout around the reactor, so neighboring channels correspond to adjacent angular positions rather than to an arbitrary listing order in the CSV. Representative series contain on the order of 6×10^5 samples at a nominal interval of $1\,\mathrm{s}$ and cover multi-day windows.

Importantly, timestamps in the *raw* CSV files are only approximately chronological: local disorders and non-monotonic segments appear, so the records are not strictly sorted in increasing time order as exported from the plant historian. The agent therefore begins with an explicit chronological sort before any downstream probe is applied.

Plant-side operational annotations and fault-related records exist for parts of this campaign; however, they are heterogeneous across sources and are not aligned into a uniform, sample-level fault taxonomy that could be tabulated without substantial manual curation. We therefore do *not* report fault-label counts in this paper. Instead, the zero-prior agent treats channel tags as opaque identifiers and does not consume fault labels during analysis; any available annotations are reserved for future supervised benchmarking rather than for the present pipeline.

For the system demonstration and case studies in section 4, we focus on a working subset drawn from Groups A1 and A2, processed independently in workspace-local analyses. The same agent protocol applies unchanged to the remaining series in the archive.

Nuclear I&C manual corpus.. The knowledge corpus comprises 59 technical documents converted from PDF to Markdown for indexing and is partitioned into a *core* pool and a *non-core* (supplementary) pool. The core pool contains 13 IAEA manuals, including the IAEA OLM reports on instrument-channel monitoring and on process/component condition monitoring and diagnostics (International Atomic Energy Agency, 2008a,b), together with related IAEA guidance on I&C design, digital modernization, maintenance/surveillance, and operational limits and procedures. The non-core pool contains the remaining 46 same-domain manuals drawn from NRC, EPRI/DOE, OECD/NEA, and related laboratory or topical materials on sensors, noise, and core/instrument monitoring. These non-core documents are *not* treated merely as noise to be rejected; rather, they are topically related supplements that enlarge the searchable knowledge base in a controlled way. A central experimental question of this work is therefore the *core–non-core relationship*: how retrieval behavior changes as the ratio of core to non-core documents varies—equivalently, how large a non-core supplement can be admitted for a fixed core set before target (core) evidence is diluted. This controlled expansion underpins the corpus-scope and self-retrieval experiments in section 3.

2.2. Zero-prior diagnostic agent

2.2.1. Design objective

The measurement-side goal is to turn a raw neutron-current CSV into a structured, human-readable condition report without assuming a labeled fault library, a fixed channel-to-physics map, or handcrafted frequency bands. Channel tags are treated as opaque identifiers; statistical morphology, redundancy, anomalies, and spatiotemporal cues are discovered from the series themselves. This "zero-prior" stance matches the archive described in section 2.1, where fault annotations are heterogeneous and are deliberately not consumed at inference time.

2.2.2. Architecture and control loop

The agent couples a local LLM planner with a deterministic tool router (Yao et al., 2023). The planner proposes tool calls; a Python router executes registered analyzers and returns concise status messages that update a persistent workspace blackboard (`state.json`). Only the entry probe may receive a raw file path; thereafter every probe must take a workspace directory alone and resolve the active CSV from the blackboard. Figure 2 summarizes the control flow.

After initialization, the planner is not allowed to restate free-text file paths. LLM tool use remains brittle: models may invent nonexistent APIs, pass malformed arguments, or generate fluent but unfaithful content (Patil et al., 2024; Schick et al., 2023; Ji et al., 2023). The workspace contract therefore keeps authoritative paths and trusted state on the runtime side, while the planner selects only the next registered probe; the failure modes that motivated these constraints are detailed in section 2.2.5.

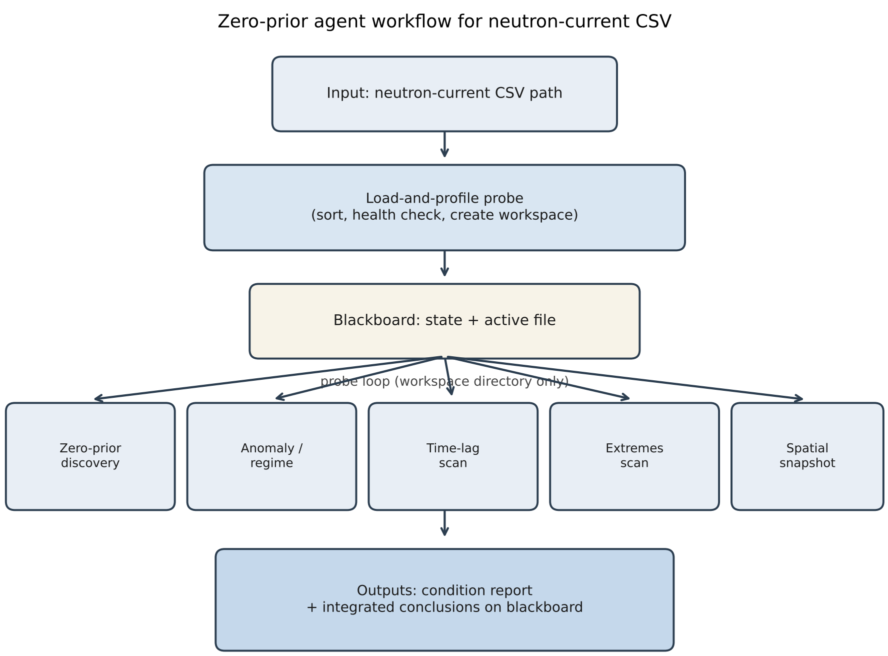

Figure 2: Zero-prior agent workflow for neutron-current CSV analysis. Profiling creates a workspace blackboard; later probes operate only on the workspace directory and accumulate a condition report.

2.2.3. Workspace initialization

Given a CSV path, the load-and-profile probe creates a dedicated workspace directory, parses the time column, sorts the series into increasing chronological order (addressing the raw non-monotonic exports noted in section 2.1), and writes a cleaned active file. It also records basic health indicators—shape, nominal sampling interval, missingness, and inactive channels—on the blackboard and appends an initialization block to the condition report. Subsequent probes refuse to run if the blackboard is missing, which enforces a mandatory warm-up step.

2.2.4. Statistical discovery and anomaly probes

After initialization, the planner may invoke the probes listed in table 2. The zero-prior discovery probe performs a global longitudinal scan: uniqueness and coefficient of variation are used to assign coarse morphological roles (discrete/step-like, high-inertia, highly volatile, or ordinary continuous), while pairwise correlations flag near-redundant channel pairs ($|r| \geq 0.99$) and strongly co-moving groups. Given the circular detector layout, such collinearity often indicates angularly adjacent or functionally redundant sensors rather than coincidental numerical agreement.

The anomaly/regime probe then segments each channel into steady and transient regimes and enumerates spike episodes and isolated zero dropouts, summarizing which channels dominate each anomaly type. The time-lag scan computes cross-correlations across channels to separate lead–lag conduction links from essentially synchronous groups. The extremes scan localizes global maxima/minima and their timestamps; the spatial-snapshot probe freezes a multi-channel cross-section at a chosen time, which is useful after an extreme has been identified. Together, these probes accumulate evidence that can later be verbalized as spikes, dropouts, redundancy, and regime changes without supervised class labels.

Table 2: Diagnostic probes used by the neutron-current agent. The last column records the registered implementation identifier for reproducibility.

Probe	Description	Implementation ID
Load-and-profile	Creates workspace, sorts timestamps, records data-health indicators.	`load_and_profile_csv_tool`
Zero-prior discovery	Infers morphological roles and co-moving or near-redundant groups.	`zero_prior_discovery_tool`
Anomaly/regime	Segments regimes; enumerates spikes and zero dropouts.	`anomaly_detector`
Time-lag scan	Estimates lead–lag versus synchronous groups.	`global_time_lag_scan_tool`
Extremes scan	Localizes global maxima/minima and timestamps.	`global_extremes_scan_tool`
Spatial snapshot	Multi-channel cross-section at a selected time.	`get_spatial_snapshot_tool`

2.2.5. Design process and adopted constraints

The architecture in section 2.2 is the outcome of an iterative, failure-driven design process rather than a single a priori specification. Early prototypes delegated almost all bookkeeping to the planner: each probe call restated an absolute CSV path, tool replies were expected to be nearly pure JSON, and the resulting condition text was passed almost unchanged into downstream diagnosis. Under a local mid-size LLM these assumptions failed repeatedly on multi-step neutron-current analyses. Table 3 summarizes the issues retained as hard constraints in the final protocol; the remainder of this subsection elaborates the failure mechanisms that motivate those rows.

Table 3: Design issues encountered in early agent prototypes and constraints adopted in the final measurement-side protocol.

Design issue	Adopted constraint
Path restatement errors across tool steps (wrong folder; raw vs. sorted copy)	Only load-and-profile accepts a CSV path; later probes take a workspace directory and resolve the active file from the blackboard.
Dropped or malformed tool JSON from local LLMs	Strip Markdown fences; extract the first brace-balanced JSON object/array; accept only registered probe identifiers.
Oversized condition text for downstream reasoning	Retain a human-readable report; drive RAG from a compact structured alert summary.
Ungrounded diagnostic prose without manuals	Build queries from the structured summary; require multi-stage, citation-bearing retrieval over the I&C corpus.

When every tool step re-emitted a free-text path, small transcription errors compounded over a probe chain: the planner confused campaign folders, mixed raw exports with chronologically sorted copies, or pointed later probes at a different series than the one just profiled. Because the runtime trusted the path argument, such mistakes executed silently and produced incoherent reports rather than explicit refusals. The adopted remedy is to treat the load-and-profile step as the sole path-accepting entry point. It creates a dedicated workspace, writes the cleaned active file, and records the authoritative location on the blackboard; thereafter probes accept only a workspace directory and resolve the series from that board (Yao et al., 2023; Patil et al., 2024). Path ownership thus moves from mutable model strings to a runtime contract, which also makes incomplete warm-up fail closed when the blackboard is missing.

A second class of failures arose at the action interface. Strict "JSON-only" prompting was brittle: local models frequently wrapped tool calls in Markdown fences, interleaved brief natural-language commentary, or truncated braces, so the router discarded otherwise valid actions; unrestricted free text made parsing equally unreliable. The present extractor therefore strips fences, searches for the first brace-balanced JSON object or array, and validates that the `action` field matches a registered probe identifier before execution (Yao et al., 2023; Schick et al., 2023). Invented analyzer names fail closed. This does not claim perfect structured generation; it treats imperfect local-model output as the default case and recovers the call when a well-formed object is present.

Even when the probe chain completed, an unfiltered multi-page condition narrative overloaded the manual-diagnosis stage: the LLM was asked to reason over lengthy prose while simultaneously selecting handbook passages, which encouraged vague, poorly grounded summaries. The design therefore separates audiences. The workspace retains a chronological, human-readable report for inspection and audit, while a compact structured summary—variable-wise alerts, spike/zero tallies, and collinearity or lag cues—is reserved as the machine interface to RAG (section 2.4). Retrieval queries are derived from that summary rather than from an operator's free-text guess about which manual clause applies, and the synthesizer is required to attach numbered citations to retrieved chunks (Ji et al., 2023). Illustrative workspaces retained for this study show complete probe chains on representative A1/A2 series, while other series exhibit partial completion or redundant re-invocation of the same probe; these cases indicate that the workspace and JSON contracts are enforceable, yet planner discipline still affects coverage—a qualitative observation that motivates the protocol, not a supervised accuracy claim with fabricated recovery rates.

2.2.6. Calling protocol and outputs

Tool calls must appear as constrained JSON objects (`{"action": ..., "parameters": ...}`), optionally wrapped in Markdown fences; the runtime extracts the first balanced JSON object or array even when the LLM adds free-text commentary, mitigating the JSON-loss issue discussed above. Only registered probe identifiers (table 2) are accepted by the local router (Ollama `qwen2.5:7b` in our deployment), so invented analyzer names fail closed rather than executing an undefined path. Each successful probe appends narrative blocks to the workspace report and updates structured fields on the blackboard (e.g., morphological roles, spike/zero summaries, lag groups). The resulting condition report—together with optional JSON snapshots—forms the sole measurement-side interface to the RAG stage in section 2.4: downstream retrieval is driven by observed phenomena rather than by an operator's free-text guess about which manual clause applies.

2.3. Manual corpus and index construction

Source PDFs are page-cropped when needed, converted to Markdown, chunked with a size/overlap grid, and embedded with `nomic-embed-text` into a LlamaIndex vector store (Liu, 2022). Indexing is applied to the corpus described in section 2.1; evaluation varies which documents are admitted into the active retrieval pool while keeping the embedding model fixed.

2.4. Report-driven multi-stage RAG diagnosis

Condition reports are parsed into compact JSON (variable-wise alerts, spike summaries, topology/collinearity cues). Queries are constructed from this JSON rather than from short free-text alone. Retrieval proceeds in stages (phenomenon/mechanism passages, then redundancy/topology supplements) and the LLM composes a multi-part diagnosis with numbered citations (`[n]`, `Rn`) linked to retrieved chunks. Because diagnosis quality depends on retrieval, section 3 evaluates the knowledge layer separately.

3. Evaluation protocol

3.1. Corpus and indexing grid

As detailed in section 2.1, retrieval experiments use a fixed IAEA *core* pool of 13 documents and a same-domain *non-core* (supplementary) pool of up to 46 manuals (59 documents in total). We vary how many non-core documents are admitted alongside the core set, thereby scanning different core–non-core ratios rather than treating non-core texts as unstructured noise. Seventeen Markdown index configurations span chunk sizes $\{800, 1000, 1200, 1500\}$ and overlaps $\{50, 80, 100, 120, 150\}$ (valid pairs only). The embedding model is fixed (`nomic-embed-text`); only the active retrieval pool and chunking configuration change across runs.

3.2. Experiment A: corpus-scope extrapolation

Four fixed queries—Chinese/English core terms and phrases (e.g., "neutron" / "neutron measurement current")—are issued while randomly attaching non-core documents in steps of five (multiple trials), so that the effective core:non-core composition changes in a controlled manner. We report top-k cosine scores, core (IAEA) purity@10 (fraction of top-10 hits from the core pool), and the rank of the first core hit.

3.3. Experiment B: self-retrieval

IAEA chunks are sampled as probes. Queries are built as first sentence, title, random window, or domain-term sentence. Relevance is scored under *strict* (same `node_id`) and *relaxed_doc* (same `doc_id`) criteria. Metrics: Recall@$\{1, 5, 10\}$ and MRR, including decay under corpus expansion and the strict–relaxed gap induced by chunk overlap.

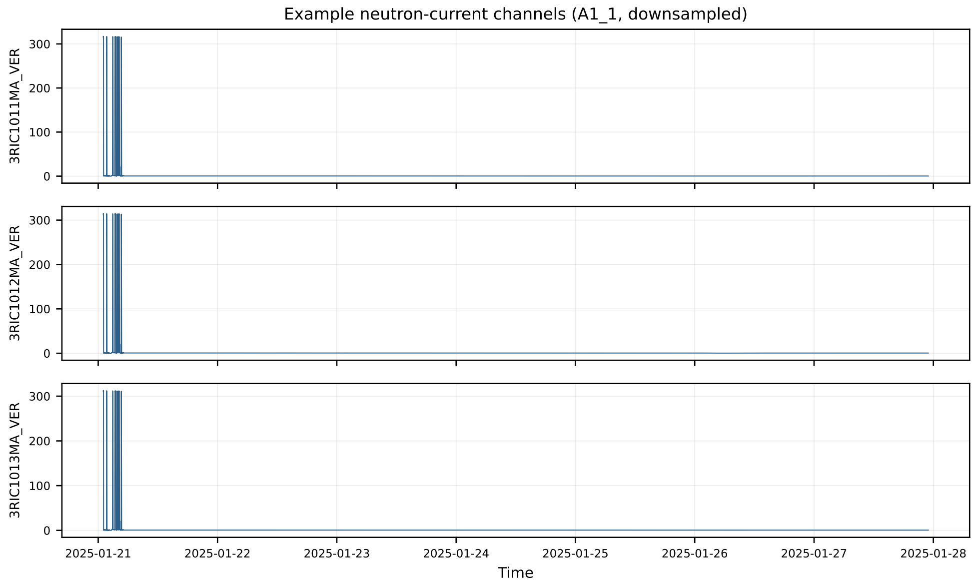

Figure 3: Example neutron-current traces from dataset A1_1 (three channels; downsampled for readability). The zero-prior agent analyzes the full-resolution series in the workspace.

4. Results and discussion

4.1. Case study: agent-to-diagnosis pipeline

On representative neutron-current workspaces (e.g., A1/A2 series), the agent produces condition reports capturing synchronized zeros/spikes and collinear channel groups; the RAG stage attaches IAEA passages on ex-core neutron noise, surveillance, and sensor characteristics to form a citable expert memo. Figure 3 illustrates three channels from A1_1 after downsampling for display. As a concrete example, the A1_1 condition report flags high-sensitivity morphology across channels, strong collinearity among several detectors, repeated spike events (notably on one channel with 23 complete spike episodes), and frequent zero dropouts on others. This subsection provides qualitative evidence that the measurement-side pipeline is operational; quantitative claims in the remainder of this section focus on retrieval.

4.2. Retrieval robustness under corpus expansion

On the IAEA-only pool, all four queries achieve perfect IAEA purity and first-hit rank 1, confirming index correctness. Under controlled admission of non-core supplements, English phrase purity declines gradually (on the order of 100% $\rightarrow\sim$34% at full same-domain expansion across configurations), while Chinese short queries often collapse near zero core purity after only about ten non-core documents—even as top-1 cosine scores rise. Thus similarity scores alone are misleading when choosing a core–non-core library ratio for deployment.

4.3. Chunking versus language

Across the chunk-size/overlap grid, full-corpus English purity typically varies only within roughly 30%–40%, far smaller than the language gap. Current bottlenecks are embedding–language alignment and query information content, not further micro-tuning of overlap.

4.4. Relevance criteria and query formulation

Self-retrieval shows a persistent gap between strict and document-level Recall@10 (e.g., about 0.65 vs. 0.86 on a representative `o80_c1200` IAEA-only setting), consistent with overlap-induced near-duplicate chunks. Domain-term sentences outperform first-sentence and random-window queries; titles may retrieve the correct document but fail strict chunk re-identification. These findings motivate the report-driven, terminology-rich query construction used in section 2.4, plus optional metadata filtering or reranking when the corpus grows.

5. Conclusions

We integrated a zero-prior tool-using agent for neutron-current diagnosis with manual-grounded multi-stage RAG, and evaluated the supporting dense-retrieval layer under same-domain corpus growth. The agent observes data; RAG aligns observations to manuals with citations; retrieval tests delineate when the knowledge layer remains trustworthy. Main empirical messages: scores can decouple from target purity; language dominates chunking; strict chunk matching underestimates useful hits when overlap is present; terminology-rich queries are preferable to titles or ultra-short Chinese prompts.

Limitations include case-level (not gold-standard) diagnosis scoring, a single embedding model, and the absence of end-to-end answer factuality metrics. Future work will add rerankers, multilingual encoders, small human-labeled query–document sets, and generation-side evaluation.

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